Striking While the Iron Is Hot vs Steady Does It: Hours Worked over the Business Cycle and Welfare

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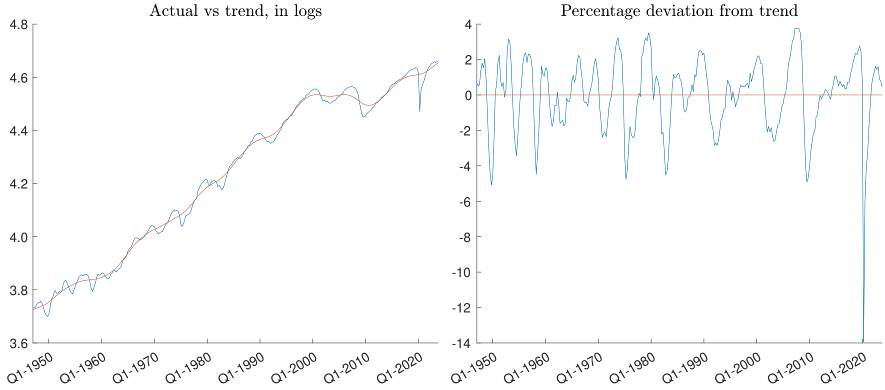
May, 2025

Introduction

- Hours worked fluctuate with productivity to raise efficiency
- But leads to costly cycles of hiring and firing
  - Loss of firm-specific capital
  - Time-consuming job search, screening, training
  - Temporary unemployment
  - Fluctuations in leisure
- **What if instead kept hours constant over business cycle?**
- Net impact -.12% of consumption, or -$66/capita in U.S. 2023
  - assuming intensive-margin adjustments, no other cost savings



Aggregate hours nonfarm business sector, U.S. 1948Q1-2023Q4, HP

Related literature

- Cho & Cooley (1994): cyclical fluct hours mostly extensive margin
- Becker (1964), Topel (1990) and Lazear (2009): firm-specific capital
- Okun (1963), Krueger & Mueller (2010) and Blatter et al (2012): search, screening and training
- Fair (1985), Fay & Medoff (1985): labor hoarding

Firms turn to layoffs to survive downturns, lack liquidity and financing

Cost of eliminating fluct in hours trivially small for society, even assuming innocuous intensive margin

Temporary wage subsidies as during COVID (Eichhorst et al. (2022))

Related literature

Cost of BC lit (Lucas (1987)) **welfare with vs without shocks**

Here, **welfare with vs without variable hours, with shocks**

Basic RBC model

$$v\left(k_0, z_0\right) = \max_{\{n_t, k_{t+1}\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t u\left(c_t, n_t\right) \tag{1}$$

subject to

$$c_t = e^{z_t} A k_t^{\alpha} n_t^{1-\alpha} + (1-\delta) k_t - k_{t+1} \tag{2}$$

$$z_t = \rho z_{t-1} + e_t \tag{3}$$

given $k_0 > 0,\ z_0$

Preferences $u\left(c_t, n_t\right)$

Risk aversion and **labor supply elasticity** crucial

- MaCurdy (1981): $\frac{c_t^{1-\varepsilon}}{1-\varepsilon} - \psi \frac{n_t^{1+1/\theta}}{1+1/\theta}$
- King, Plosser, Rebelo (1988): $\frac{c_t^{1-\varepsilon}\left(1-\frac{\psi(1-\varepsilon)}{1+1/\theta} n_t^{1+1/\theta}\right)^{\varepsilon}-1}{1-\varepsilon}$
- Greenwood, Hercowitz, Huffman (1988): $\frac{\left(c_t-\frac{\psi}{\omega} n_t^{\omega}\right)^{1-\varepsilon}-1}{1-\varepsilon}$

RRAC for consumption $\left(-u''_{cc} c/u'_c\right)$ is ε

Frisch elasticity of labor supply is θ or $1/(\omega-1)$

MaCurdy yields largest impact, so focus on that

Solving model

- Discrete grid search over capital k
- Use FOC to substitute out for n, or $n = n_{ss}$ when constant
- 51 gridpoints for exogenous productivity z
  - Cost of constant hours decreasing in gridpoints for z
  - Fairly stable with 31 or more gridpoints for z
- 4 000 or 8 000 gridpoints for capital k
  - Cost of constant hours decreasing in gridpoints for k
  - Fairly stable with 2 000 or more gridpoints for k

Calibration

- Frisch labor supply elasticity $\theta = 2$
- Capital share $\alpha = 1/3$
- Discount $\beta = .99$
- Depreciation $\delta = .025$
- Persistence $\rho = .9$
- RRAC $\varepsilon \in \{0, .5, 1, 2, 3, 4, 5, 10, 15, 20\}$
- Calibrate variance of $z-$shocks so HP-filtered c and n at least as volatile as in U.S. data

Welfare cost of eliminating cyclical fluctuations in hours

RRAC									
0	.5	1	2	3	4	5	10	15	20
.04	.09	.07	.12	.25	.52	.90	2.61	4.50	6.35

Table: Welfare cost of eliminating cyclical fluctuations in hours as a percentage of steady-state consumption. MaCurdy preferences.

Meta-analysis by Elminejad et al. (2022): 1 021 estimates from 92 studies, yields mean RRAC of 1

Chetty (2006): RRAC > 2 inconsistent with observed labor supply

Robustness

	α	β	δ	θ	ρ
Low	(.25)	(.985)	(.01)	(.5)	(.85)
	.10	.09	.25	.18	.16
Baseline	(1/3)	(.99)	(.025)	(2)	(.9)
	.12	.12	.12	.12	.12
High	(.45)	(.995)	(.04)	(3)	(.95)
	.17	.20	.07	.14	.16

Table: Welfare cost of eliminating cyclical fluctuations in hours, % c_{ss} for different parameter values. MaCurdy, RRAC $= 2$.

- $\varepsilon = 2$, $\alpha = .45$, $\beta = .995$, $\delta = .01$, $\theta = 3$, $\rho = .85$ yields .47% or \$264/capita in 2023

Forcing hours $n = n_{ss}$

- Raises $\bar{n}$ for RRAC ≥ 1, lowers $\bar{n}$ for RRAC $\leq .5$
- Raises $\bar{c}$ for RRAC ≥ 2, lowers $\bar{c}$ for RRAC ≤ 1
- Raises σ_c for RRAC ≥ 2, lowers σ_c for RRAC ≤ 1
- For RRAC $= 1$: $\bar{n} + .07\%$, $\bar{c} - .16\%$, $\sigma_c - 19.3\%$
- For RRAC $= 2$: $\bar{n} + .27\%$, $\bar{c} + .15\%$, $\sigma_c + 15.7\%$

Different preferences

	RRAC									
Pref	0	.5	1	2	3	4	5	10	15	20
MaC	.04	.09	.07	.12	.25	.52	.90	2.61	4.50	6.35
KPR	-	.04	.08	.04	.05	.05	.05	.06	.07	.07
GHH	.04	.08	.09	.09	.09	.10	.10	.11	.12	.14

Table: Welfare cost of eliminating cyclical fluctuations in hours, % c_{ss}. MaCurdy, KPR and GHH preferences.

As % of $\bar{c}$ impact is slightly lower, but similar

	RRAC									
	0	.5	1	2	3	4	5	10	15	20
MaC	.04	.09	.07	.12	.25	.52	.90	2.61	4.50	6.35
KPR	-	.04	.08	.04	.05	.05	.05	.06	.07	.07
GHH	.04	.08	.09	.09	.09	.10	.10	.11	.12	.14

Table: Welfare cost of eliminating cyclical fluctuations in hours, % c_{ss}.

	RRAC									
σ	0	.5	1	2	3	4	5	10	15	20
MaC	.015	.049	.061	.085	.101	.118	.131	.138	.140	.140
KPR		.048	.061	.043	.041	.040	.040	.038	.037	.036
GHH	.015	.033	.033	.033	.033	.033	.033	.033	.033	.033

Table: Volatility σ necessary for simulated SD of HP-filtered consumption and hours to be at least as high as in U.S. data.

Overestimate cost of constant hours

- Assume innocuous intensive-margin adjustments, rather than costly hiring and firing
- Model economy calibrated to be at least as volatile as U.S.
- Only source of volatility is productivity
- n_{ss} may not be optimal constant level of hours
- Finite number of gridpoints

Underestimate cost of constant hours?

Model changes requiring more volatile shocks raise cost estimates!

Already unusually high compared to volatility of Solow residuals

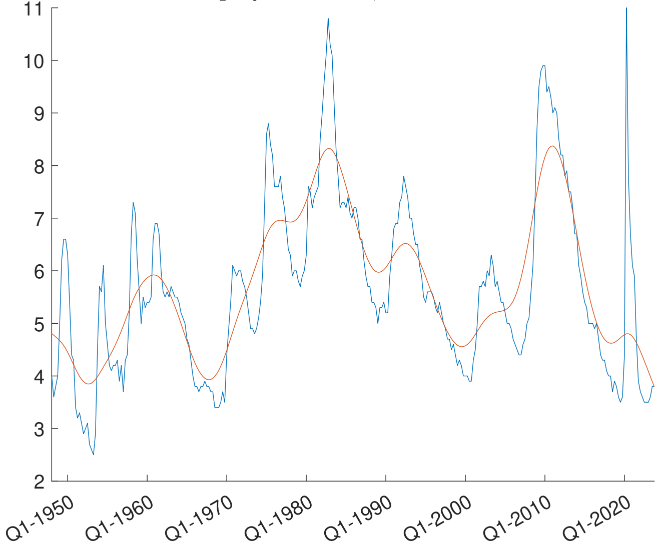
But even doubling calibrated volatility, cost at most .5% for standard parameter values (RRAC ≤ 2) across 3 utility functions

Conclusion

Cost of eliminating fluct in hours trivially small, even assuming innocuous intensive margin

- $<$ .12% or \$66, standard parameter values, 3 utility functions
- $<$ .5% or \$264, extreme parameters values or doubling volatility

Unemployment rate, actual vs trend



Substituting out for n

Use FOC for n_t,

$$\psi n_t^{1/\theta} c_t^{\varepsilon} = (1-\alpha)\, e^{z_t} A k_t^{\alpha} n_t^{-\alpha} \tag{4}$$

together with

$$c_t = e^{z_t} A k_t^{\alpha} n_t^{1-\alpha} + (1-\delta) k_t - k_{t+1} \tag{5}$$

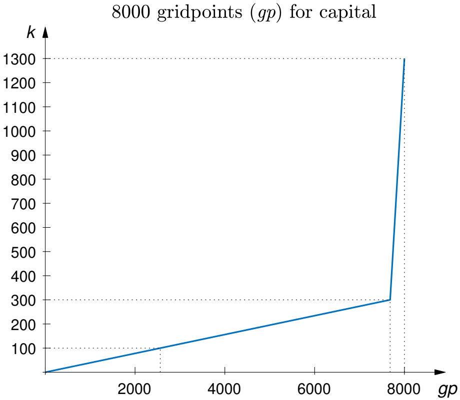
to solve numerically for optimal n_t for each of the 816 million possible combination of z_t, k_t and k_{t+1}

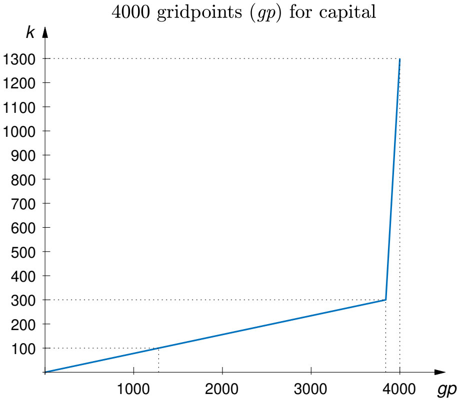
Gridpoints n determined by FOC and gridpoints k and z

Gridpoints capital k

- 96% equally spaced out (linearly) $[0, 3k_{ss}]$; .078; .039
- 4% equally spaced out $[3k_{ss}, 13k_{ss}]$; 6.329; 3.145
- $k_{ss} = 100$; $gp = 4\,000$; $gp = 8\,000$

Gridpoints capital k





	RRAC									
gpz	0	.5	1	2	3	4	5	10	15	20
101	.038	.086	.074	.118	.255	.524	.903	2.60	4.48	6.32
91	.038	.086	.074	.118	.255	.524	.903	2.60	4.49	6.33
81	.038	.086	.074	.118	.255	.525	.904	2.60	4.49	6.34
71	.038	.086	.074	.118	.256	.525	.905	2.60	4.50	6.34
61	.038	.086	.074	.118	.256	.527	.907	2.61	4.51	6.36
51	.038	.086	.074	.119	.257	.529	.910	2.62	4.52	6.38
41	.039	.087	.075	.119	.259	.532	.916	2.64	4.55	6.42
31	.039	.088	.076	.121	.262	.539	.928	2.67	4.62	6.51

Table: Welfare cost of eliminating cyclical fluctuations in hours, % c_{ss}, computed with *gpz* gridpoints for z. MaCurdy preferences, $gp = 2\,000$.

	RRAC									
gp	0	.5	1	2	3	4	5	10	15	20
4 000	.038	.086	.074	.117	.254	.524	.905	2.61	4.50	6.35
2 000	.038	.086	.074	.119	.257	.529	.910	2.62	4.52	6.38
1 000	.038	.087	.077	.125	.268	.544	.932	2.67	4.61	6.50
500	.038	.090	.086	.151	.312	.608	1.02	2.88	4.95	6.99

Table: Welfare cost of eliminating cyclical fluctuations in hours, % c_{ss}, computed with gp gridpoints for k. MaCurdy preferences, $gpz = 51$.